



11166 - Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with PRISM Phase Curves

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	ZTFJ1828+2308	NIRSpec Bright Object Time Series	(1) ZTFJ1828+2308
	2	ZTFJ1230-2655	NIRSpec Bright Object Time Series	(2) ZTFJ1230-2655

ABSTRACT

Spectroscopic phase curves offer the most direct and informative observational constraints of key atmospheric processes in irradiated brown dwarfs. We propose NIRSpec PRISM BOTS observations of two low mass white dwarf-brown dwarf binaries, ZTFJ1828+2308 and ZTFJ1230-2655, bridging the gap between higher mass irradiated brown dwarfs and lower mass hot Jupiters. We will use high signal to noise PRISM observations to obtain 0.6-5.3 micron phase-resolved spectra. White dwarf-brown dwarf binaries are particularly advantageous for these observations due to their short orbital periods and improved contrast between the white dwarf and the brown dwarf in the infrared.

These phase curves will allow us to probe the atmospheres of some of the lowest mass objects that survive a violent phase of common envelope evolution. Retrieved chemical abundances will give insights into the formation history of these objects, and the imprints that common envelope evolution has left on their atmospheres. Our observations will allow us to constrain energy transport rates with model-independent temperature contrasts, obtain 3D temperature and cloud maps of the brown dwarfs, derive molecular abundance maps to use as probes of quenching, and understand the fundamental properties of post-common-envelope brown dwarfs. Our analysis of the phase curves of ZTFJ1828+2308 and ZTFJ1230-2655 will characterise their atmospheres in detail and identify atmospheric tracers of common envelope evolution.

OBSERVING DESCRIPTION

We will conduct NIRSpec/PRISM Bright Object Time Series (BOTS) observations of two low mass white dwarf-brown dwarf binaries. We aim to obtain phase curves in emission of the brown dwarfs from 0.6-5.3 microns.

Before spectroscopic observations for each target, one set of acquisition exposures will be taken. The observations will be taken using the SUB512 array with the NRS readout pattern. The groups per integration numbers are set to maximize the SNR and ensure good cadence, and the total integration number is set to ensure the target is monitored for one period.

These observations result in a total of 13.2 hours of charged time and 9.6 hours of science time. Our ETC runs predict that the SNRs we will reach are sufficient for our science goals.

We will observe ZTF J1828+2308 with 92 groups per integration for a total of 140 integrations, yielding a total exposure time of 11686.293 seconds.

We will observe ZTF J1230-2655 with 115 groups per integration for a total of 220 integrations, yielding a total exposure time of 22941.653 seconds.

As we are observing full orbits of each target, we do not require a phase constraint.

Proposal 11166 - Targets - Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with PRISM ...

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ZTFJ1828+2308	RA: 18 28 48.7736 (277.2032233d) Dec: +23 08 38.39 (23.14400d) Equinox: J2000	Proper Motion RA: -6.134 mas/yr Proper Motion Dec: -19.492000001264387 mas/yr Parallax: 0.0049145000000000005" Epoch of Position: 2000	
Fixed Targets	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Eclipsing binary stars, White dwarfs]			
	(2)	ZTFJ1230-2655	RA: 12 30 16.5915 (187.5691312d) Dec: -26 55 51.34 (-26.93093d) Equinox: J2000	Proper Motion RA: -41.48 mas/yr Proper Motion Dec: -7.893000019976171 mas/yr Parallax: 0.0051128" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Eclipsing binary stars, White dwarfs]				

Proposal 11166 - Observation 1 - Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with P...

Observation	Proposal 11166, Observation 1: ZTFJ1828+2308										Fri Mar 13 21:02:29 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpec Bright Object Time Series											
Diagnostics	(ZTFJ1828+2308 (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	ZTFJ1828+2308	RA: 18 28 48.7736 (277.2032233d) Dec: +23 08 38.39 (23.14400d) Equinox: J2000			Proper Motion RA: -6.134 mas/yr Proper Motion Dec: -19.492000001264387 mas/yr Parallax: 0.004914500000000005" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star Description=[Brown dwarfs, Eclipsing binary stars, White dwarfs]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	275183
Template	Subarray										
	SUB512										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRS	92	140	1	1	140	11686.293	275183	

Special Requirements	Time Series Observation No Parallel Attachments
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Proposal 11166 - Observation 2 - Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with P...

Observation	Proposal 11166, Observation 2: ZTFJ1230-2655										Fri Mar 13 21:02:29 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSspec Bright Object Time Series										
Diagnostics	(ZTFJ1230-2655 (Obs 2)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	ZTFJ1230-2655	RA: 12 30 16.5915 (187.5691312d)			Proper Motion RA: -41.48 mas/yr					
			Dec: -26 55 51.34 (-26.93093d)			Proper Motion Dec: -7.893000019976171 mas/yr					
			Equinox: J2000			Parallax: 0.0051128"					
						Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[Brown dwarfs, Eclipsing binary stars, White dwarfs]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	275183
Template	Subarray										
	SUB512										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRS	115	220	1	1	220	22941.653	275183	

Special Requirements	Time Series Observation No Parallel Attachments
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